CSE 111

Section : 07 (Sun - Tues - 11:00 TAW)

13/10/2025

String

1. s1.length()
2. s1.charAt()
3. s1.codepointAt() → most use in loops
4. (int) char
5. (char) int
6. Concatination
7. s1.equals()

String is immutable (.replace create new string)
Array is mutable

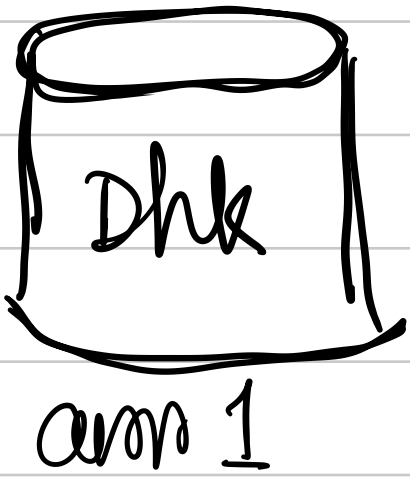
Arrays

① arr1.length

② arr1[index]

Array storing

int [] arr1 = new int [4]



arr 1

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(OOP)

Blueprint / Design / Template → Product

↓ ↓

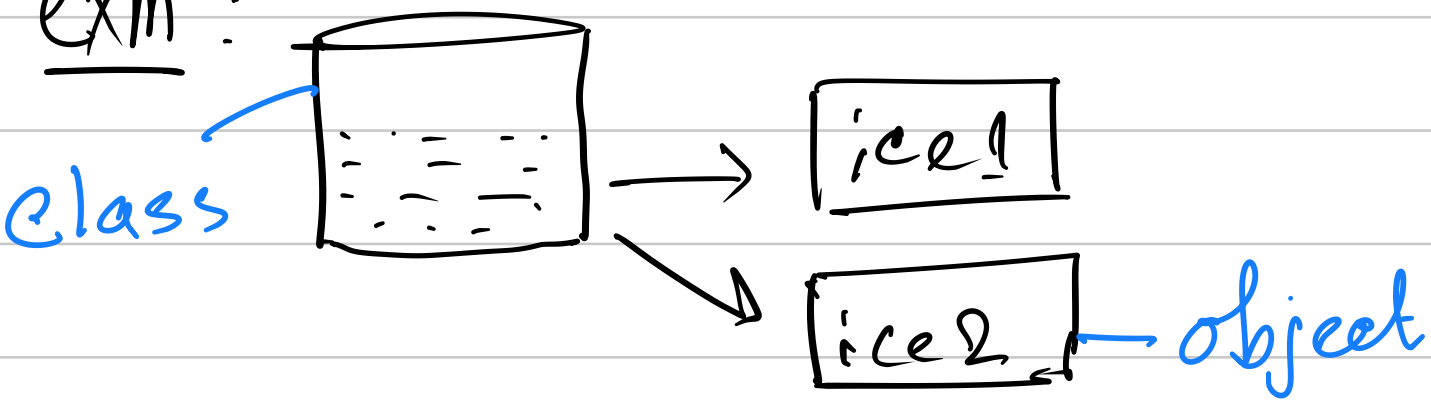
class → objects/instance

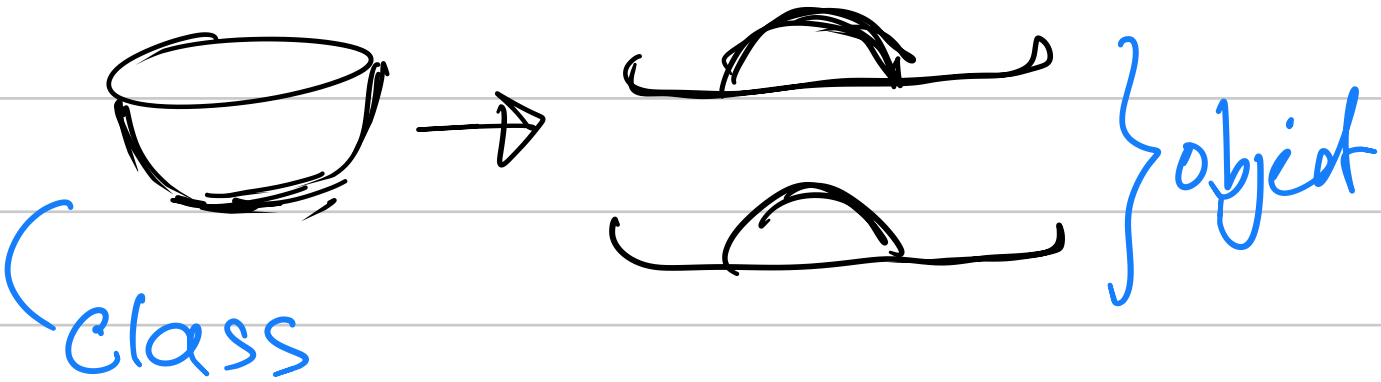
↓

User defined datatype.

* objects are independent

exm :





Never even two object in a same location that why they are independent.

* Designs are non-physical.
* objects → Data (variable)
  → Behavior (Method)

* We can make as much as we want using the one class.
* Classes are user defined datatype.
* //classname variable =
  new classname( )
  → object / instance

* Student s1 = new Student ( );

Datatype of s1

sout(s1)

// Student@54dfe → memory location

↓

classname



DHK :



S1

19|10|2025 (class-3)

①

Variable	instance variable
regular variable	depend on object class

②

design class	tester/driver class
	access the design class.

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Variable = 3 (s1, s2, s3)

```
Student s1 = new Student();
Student s2 = new Student();
Student s3;
    s3 = s2;
```

object creation = 2

* object will only create after (new) keyword.

```
// access_modifier void/return-type methodName (parameters){
                                                            }
```

* Method will not worked until it get called.

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Method Overloading

↓

Ekta class e same name er method.

↙ different number of parameters

↘ different datatype

S1 = Dhaka

name	id	uni	UpdateName()	ShowDetails()	addcource() c1

addcource c1	c2	addcource() int

An array always come with a count variable must.

```
public String[ ] course : new String[5]
public int count = 0 ;
```

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Constructor

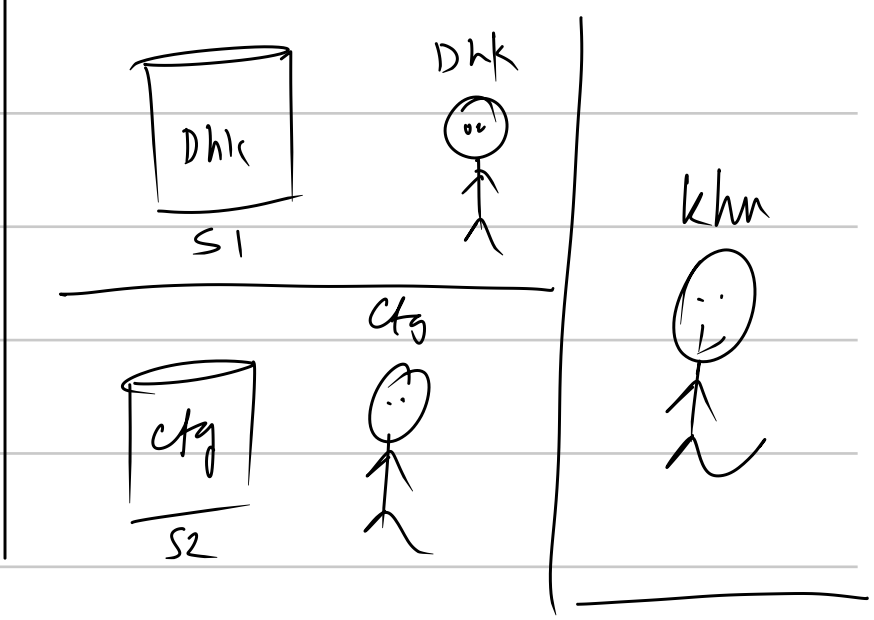
* a special default method()
* Per object - constructor called once
* object can't be creat without calling this
* Every time we use constructor it creats in different locations.

```
//access_modifier  ClassName(Parameter)
```

* It will never return any value.

method	constructor
* void/return থাকবে	* থাকবে না
* ইচ্ছামত নাম	* class এর নাম
* multiple time called	* একবার

Student s1 = new Student( )
Student s2 = new Student( )
new Student( );



Pass by Reference

int → 5
double → 8.5
boolean → true
String → "CSE"
} pass by value

Student → s2 → pass by ref.

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How to appoch: (Assignment 4) Task-01

Triangle
- a
- b
- c
- perimeter
} Veriable

- Constructor ( )
- void triangleDetails( )

return ├ printTriangleType ( )
void ├ compareTriangle (Triangle) } method

tricks

```
int method = 6;
  method = 6+2;
s.o.p (s1.method);
```

{ printing details ↳ only print,
no calculation

```
int[] A = {2,7,6}
public void test(int [] a){
```

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Static

Static variable doesn't dependent on object.
we can access it directly. It called directly after the design class name.

Static Method

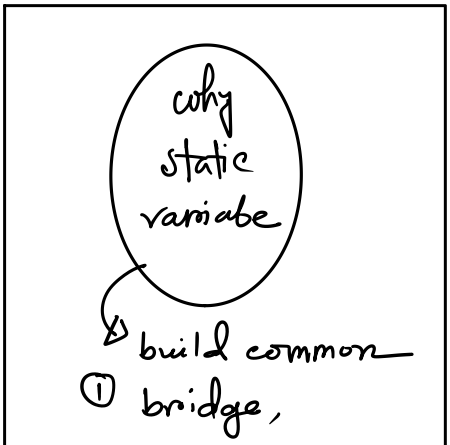
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Instance

- Instance variable is object dependent

static

- multiple objects share same variable
- main method is also a static method.
- object independent.



object ผ่าน reference
↳ call ด้วย method
↳ instance method

class ใช้ ref ↳
call ด้วย method
↳ static method.

② without object dependent

③ Do not count individual.

↳ s1.upUniName (Brac University)

static method no need to put in every table,
→ Common property, common for everyone.
→ accessible for everyone.

instance method for instance varible ⇒ static method for static variable

```
public static void upUniName(){
    student.uniname = uname; }
```

```
Student.upUniName(" ")
```

public / private variable.

private variable
- setName()
- getName() → this method will always return thing.

(↳ private method must have 2 method. (set and get)
↳ private variable accessible in the design class.)

Instance Method
- → Instance Variable
- → Instance Method
- → static variable
- → static Method

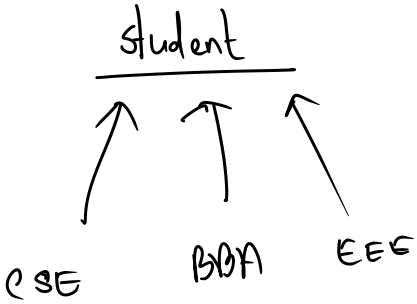
} all can do

Static Method
- → Static Variable
- → Static Method
- → No
- → No

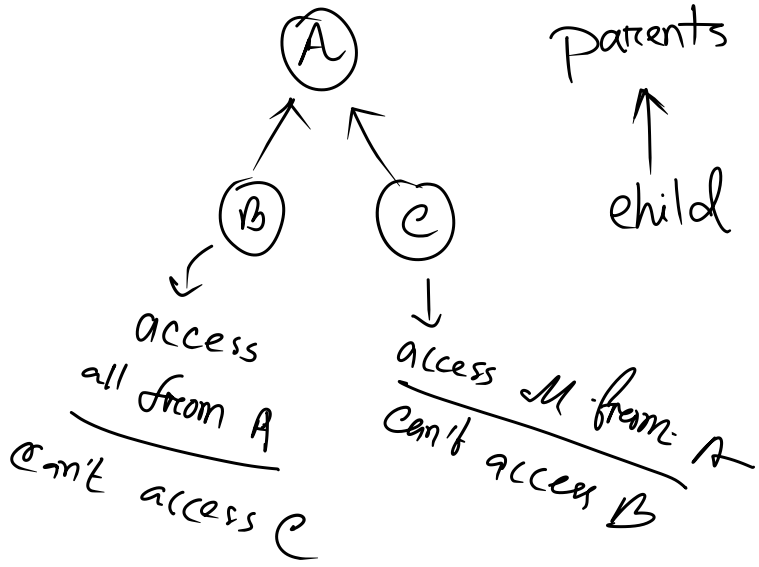
They are object oriented. / a static block can not access any unstatic block.

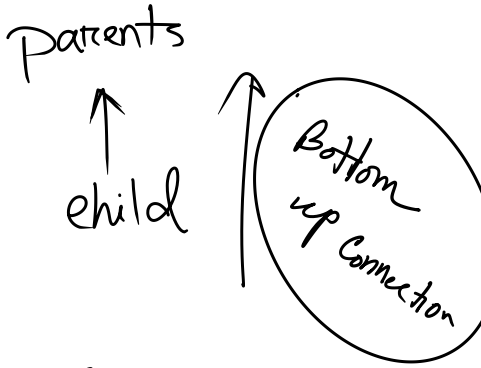
Inheritance

oparents doesn't know how many child they have.



IS-A relation



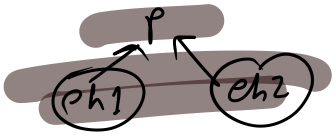


Type of Inheritance

Parent ← child → Single

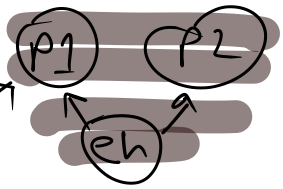
P ← ch1 ← ch2 → Multilevel

→ Hiarcy

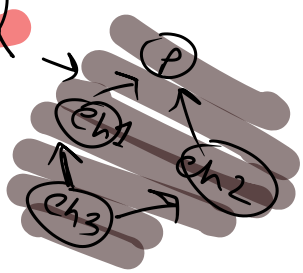


not in java

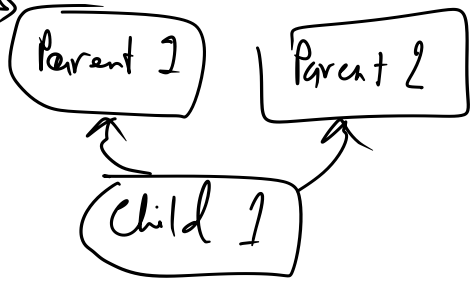
→ Multiple

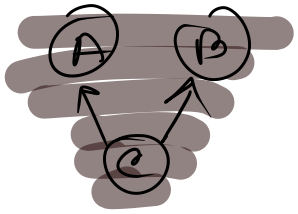


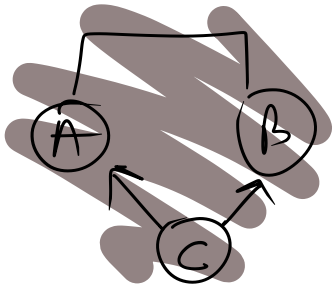
→ Hybrid



Multiple Inheritance can not support java



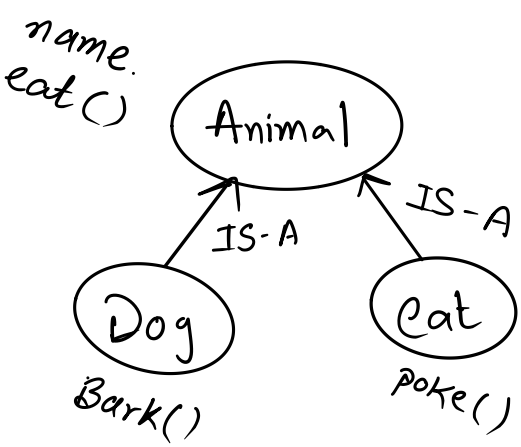




* Diamond problem in Java.

↳ creats a cycle of parents and can not access properly so its a diamond problem

14/12/2025



Inheritance (Parent - child)

(Super – sub)



A is super class of B

B is super class of C

```
public class Dog extends Animal { }

public class Animal {}
```

when child class written there is a build in method to call the parent class constractor. `super();`

```
public class Dog extends Animal {
    public Dog() {
        super();
        sop() }
    }
```

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toString( )

```
publis String toString(){
        return "I am a cat";
      }
Cat c1 = new Cat("Tom")
 S.O.P (c1)

// I am a cat.
```

object class la toString( ) method.
default.

toString( ) always return string. Not any other data types.



if I called e then first called will A's constructor then B

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Polymorphism

Poly | Many

Morph | Forms

a single thing in different forms

Polymorphism

compile time

Fixed Binding
(method overloading)

when we do method overloading we write same method with different perameter and in compile time it decide which method will be called. That's why it's compile time polymorphism.

-> compiler check left side.

-> compiler check Reference

Run Time

Dynamic Binding
(method overriding)

- Run time check Right side
- Run time check object /class
- We need to write the same method to every class.
- when we do method overriding we write the exact method in difference class and in Run time we check which class object will be called. That's why it's Run Time polymorphism.

Reference Type var = new object ().

- Child class's object can be store in parent's object.
  parent class can not.

```
Animal a2 = new dog()
Dog DI= new dog()
Animal aI = DI
SOP(DI) // Dog@
```

explicit casting double x =(double) 7

implicit casting double x = 7

(char) 66

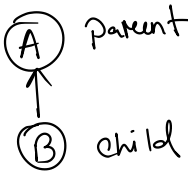
(int) 'A'

(d1 instanceof Dog) // true

(d1 instanceof Animal) // true

instanceof

downcasting & Upcasting



convert child to parents → upcasting

convert parent to child → downcasting

left side ला जर्सर Referrence type.

Right side ला object

chocolate choco = new chocolate()

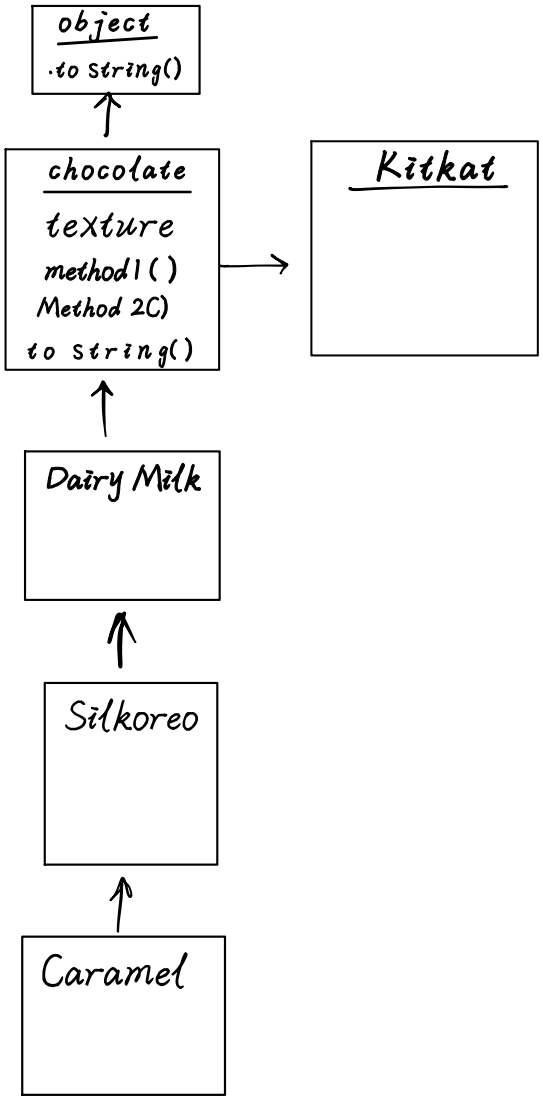
Animal a = new dog();

Dog d1 = (Dog) a

```
public static void display(animal a){ // upcasting
    a.makesound()
}
public static void main(String[] aregs){
    Dog d1 = new dog();
    display(d1);
}
```

যার parent নাই তার parent object class and object class এর
কোনটি method .toString()

30/12/2025 : Tracing



method called check right side / object
variable called check left side / refferene
└ only check refference

Reference object

((silkorea). dairy milk) method 3 ()

casting type

method will always called based on object

#Error is also one kind of output

⇒ compile time
⇒ Run time

Java behave differently for variable and method

variable maintain scope
method check ownself first.